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MICROBIOLOGY FOR NURSES



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MICROBIOLOGY AND THE NURSE

Microbiology is essential to the nurse students nurses need to apply knowledge of microbiology to every day professional practice.

In every field of Training and patient care a knowledge of basic principles of microbiology is essential. This knowledge will insure correct procedures in handling babies, in assisting at operating room, in changing surgical dressings and in caring for patients with communicable diseases.

A nurse must actually has a highly important role to play in each aspect of the management of infectious diseases; in diagnosis, treatment and prevention. She is going to participate in making the diagnosis through accurate reporting of patient's symptoms and by assuring well-timed careful collection of specimens ordered for laboratory studies to identify the cause of infection. Using her knowledge of the nature of microbial agents of infectious diseases, the nurse must apply appropriate precautionary measures in her bed-side care of every patient so that communicable diseases are prevented from spread from one to another. The clever student soon recognizes the importance of learning the principals of microbiology and its applications in medicine.

PART I

BASIC PRINCIPLES OF MICROBIOLOGY

General Characters of Microorganisms

Microbiology is the subject that deals with the study of organisms which are too small to be seen with the naked eye. The micro-organisms that will be studied in this course are bacteria and viruses.

Bacteria: are unicellular organisms. Their size are measured in microns ($1\mu = 1/1000 \text{ mm}$). Each bacterium is a functioning living unit, deriving its food from the surroundings and uses this food to manufacture the complex materials necessary for life and reproduction.

Bacteria are widely distributed in nature, and according to their habitat and way of living they are described as:

- 1) Saprophytic.
- 2) Parasitic.

Saprophytic bacteria live on inanimate material; in the soil, in water, in dust, in the air, on clothes or on dead bodies and decaying organic matter.

Parasitic bacteria live on or in the body of living creatures.

Parasitic bacteria are classified into:

- 1- Pathogenic.
- 2- Commensals.

Pathogenic bacteria: cause disease in man or animals or plants, while commensals bacteria : live on or in the body without exerting a harmful effect. Some of these commensals may be potentially pathogenic and may cause disease if the body resistance is lowered by any means.

Morphology of Bacteria:

1- Size of the Bacterial Cell:

The majority of bacteria fall within the general dimensions of 0.75 to 4 μm .

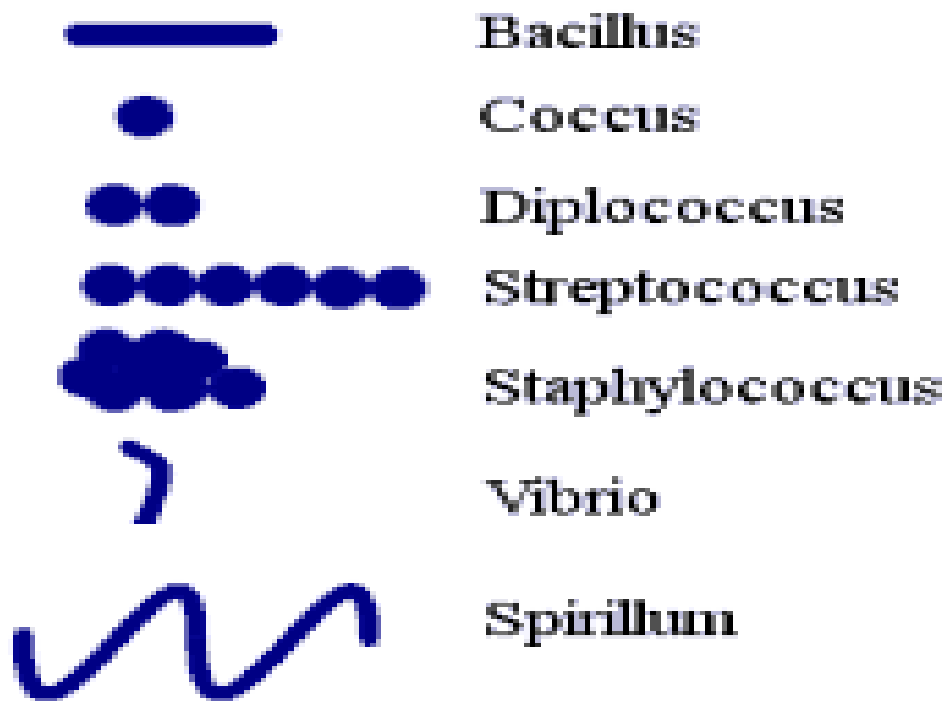
2- Shape of the Bacterial Cell:

They are unicellular structures which may occur as

- Bacilli (rod-shaped) e.g. the typhoid bacillus ,
- Cocci (spherical shape) e.g. Staphylococci ,
- Vibrios (comma shape) e.g. V. cholera,
- Spiral shape e.g. spirochaetes.

3- Arrangement:

When the bacterial cell divides, the two daughter cells may at once separate, or they may remain attached to one another by the cell membrane. It may occur in clusters (e.g. Staph) some in pairs (diplococci) e.g. (Pneumococci) and others occurring in chains e.g. (Streptococci), (Figure:1).



Figure(1): Bacterial Morphology

Structure of the Bacterial Cell

A-(Essential structure):

1- Cell Wall:

- **Gram Positive Bacteria Cell Wall:**
 - ❖ **Peptidoglycan layer:** which is unique structure in bacteria. It forms about 50% of the cell wall material. It gives rigid support, and its synthesis inhibited by penicillins and Cephalosporins.
 - ❖ **Teichoic acid layer:** Major surface antigen and important for attachment of bacteria to mucosal cells.

- **Gram Negative Bacteria Cell Walls:**

- ❖ **Lipopolysaccharides (LPS)** : is the outer membrane of the cell wall of Gram-ve bacteria. It is endotoxin and responsible for septic shock(fever, hypotension and shock).
- ❖ **Peptidoglycan layer**: inner layer , only 5-10% of cell wall material.

Function of the Cell Wall:

- 1- It's a rigid structure that maintains shape of bacteria.
- 2- Protects the cytoplasmic membrane from bursting in hypotonic solutions.
- 3- Detects the response to Gram stain.
- 4- Play a role in cell division.

2- Cytoplasmic membrane: It is a semipermeable double layered structure lies inside the peptidoglycan layer of the cell wall. It is composed of phospholipid bilayer.

The important Functions :

- 1- Active transport of nutrients into the cell and waste out of the cell (Selective permeability).
- 2- Play a role in respiration and cell division.
- 3- Secretion of enzymes and toxins.

3- Bacterial Cytoplasm:

The cytoplasm is a soft gel containing a variety of numerous granular structures.

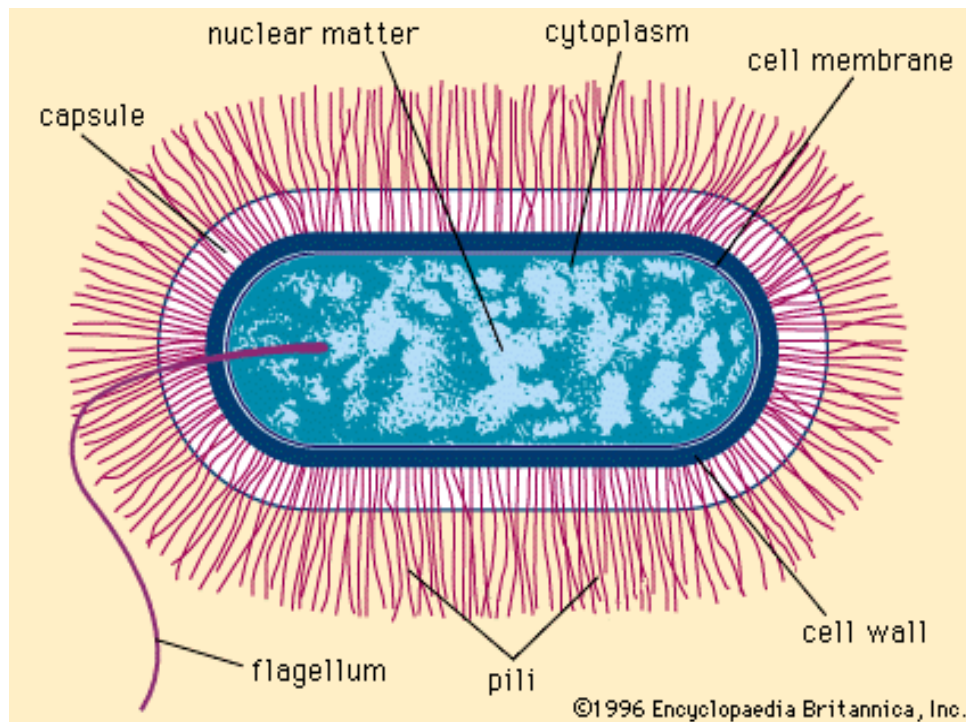
- (1) Fat granules and polysaccharide granules.
- (2) Glycogen granules.

(3) Ribosomes Concerned with protein synthesis.

4- Bacterial nucleoid: Within the cytoplasm there is a structure that carries on the functions of a nucleus especially hereditary functions and control of activities concerned with cell growth and multiplication. This nuclear material of deoxyribonucleic acid (DNA) is not bounded by a nuclear membrane as occurs with animal and plant cells and does not have any nucleolus (Figure 2).

B-(Non-essential structure):

- 1- **The capsule:** a protective gelatinous layer outside the cell wall. It is like a jacket, a virulence factor of many bacteria as it protects bacteria against phagocytosis.
- 2- **Flagella:** These are long, delicate, appendages attached to the bacterial cell. These projections can be seen by special methods. They are composed of protein (flagellin) that is antigenic. It is concerned with motility.
- 3- **Pilli (Fimbriae):**
These are very thin and short thread-like structures found on the surface of certain bacterial cells. They are present in Gram-negative bacilli and responsible for adhesion & attachment to mucosa.



Figure(2): Bacterial Structure

4-Spores:

The highly resistant structures are formed by two medically important genera of gram positive bacilli, the genus *Bacillus* and the genus *Clostridium*.

Spore formed under unsuitable environmental conditions when nutrients as carbon and nitrogen are depleted. The spores are formed inside the cell and contain bacterial DNA, small amount of cytoplasm, cell membrane, peptidoglycan and a thick keratin like coat responsible for resistance to heat, dehydration, radiation and chemicals.

They are like seeds of plants which survive without food or water for months or even years, but grow and mature again when a suitable temperature, the food and water essential for life become available.

Biological Needs of Bacteria:

For bacteria to grow and multiply, certain requirements must be met which are:

- 1- Bacterial Nutrition: Sufficient food which must contain all the required ingredients and essential metabolites.
- 2- An adequate level of moisture.
- 3- Suitable degree of temperature.
- 4- A proper degree of alkalinity or acidity.
- 5- The proper requirements of oxygen according to the nature of bacteria.
- 6- Absence of light completely or partially.
- 7- Avoid the accumulation of the bi-products of bacterial growth in the medium.

Significant deviation from any of these requirements will greatly affect the growth of the bacterial population under investigation:

1- Nutritional Requirement of Bacteria:

As mentioned before, there is variation in the requirement of nutritive materials needed by different bacterial cells. This variation depend on the nature of the bacteria and its type whether from the heterotrophic group or from the autotrophic one.

I- Autotrophs: These are bacteria which can utilize simple inorganic substances, e.g. CO₂ as a source of carbon and ammonium salts as a source of nitrogen; from which they synthesize organic substances, e.g. proteins, carbohydrates etc...

The energy needed for their metabolism is obtained from light or from oxidation of inorganic substances. These are free living, non-parasitic organisms of no direct medical importance.

II- Heterotrophs: These bacteria require complex preformed organic substances, e.g. sugars, proteins etc..., which are derived from plant or animal sources. All bacteria of medical importance are heterotrophs. They live in or on the animal body and are called parasitic bacteria. Many of the heterotrophs can grow on simple media containing peptone or meat extract as in broth and nutrient agar. Others require complex organic material, e.g. blood or serum. Media containing such substances are called enriched media.

2- Moisture:

Water is essential for bacterial growth as it acts as a vehicle for food transmission. Drying may affect some types of bacteria while others can resist drying for different periods according to their degree of resistance. So we find spores resist dryness while delicate bacteria as gonococcus resist dryness only for few hours.

3- Temperature:

All the processes of growth depend on chemical reaction. The rate of these chemical reactions is influenced by temperature. Each species of bacteria will grow at a temperature within a certain range and hence bacteria can be classified according to their ability for growth at different degree of temperature as follows:

- 1- Psychrophiles: which can grow at 0°C or lower. They can also grow at higher temperature. Optimal growth takes place between 20-30°C.
- 2- Mesophiles: which grow best within a temperature range from 25-40°C.

3- Thermophiles: which grow best at temperature between 45°C to 60°C some types of this category grow above 60°C and cannot grow below this degree. These strains are known as obligate thermophile or sterothermophilus. The growth range of some thermophilic bacteria extends into the mesophilic group. These species are known as facultative thermophiles or enrithrophiles.

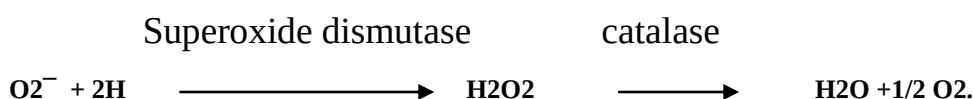
The temperature of incubation which allows for most rapid growth during a short period of time (12-24 h.) is known as the optimum temperature. This optimum temperature differs from one species to another.

4- Acidity or Alkalinity (PH):

For most bacteria the optimum pH for growth is between 6.5-7.5. Most organisms can grow at a pH between 4-9. As a result of bacterial growth, there is change in pH due to the excretion of organisms which may be acidic or basic. These changes of pH may inhibit the growth of organism.

5- Gaseous Requirements:

For most organisms, an adequate supply of oxygen enhances metabolism and growth. The oxygen acts as the hydrogen acceptor in the final steps of energy production catalyzed by the flavoproteins and cytochromes. The use of oxygen of oxygen generates 2 toxic molecules, hydrogen peroxide (H₂O₂) and the free radical superoxide (O₂⁻), bacteria requires 2 enzymes to utilize oxygen the first is superoxide dismutase and the second is catalase



A- Oxygen: According to their oxygen requirement, bacteria are classified into 4 groups:

1- Obligate aerobes: Some bacteria will only grow in the presence of oxygen because their energy generating system is dependent on oxygen as the hydrogen acceptor, e.g. *M. tuberculosis* and *Pseudomonas aeruginosa*.

2- Facultative anaerobes: These bacteria can grow in the presence or absence of O₂. They use O₂ to generate energy by aerobic respiration if it is present. However, they can use anaerobic respiration to generate energy in the absence of O₂.

3- Obligate anaerobes: These bacteria grow only in the complete absence of O₂ and die in its presence, e.g. *Cl. tetani*. In presence of O₂, two toxic molecules are produced; hydrogen peroxide and superoxide ion. Anaerobic bacteria lack the enzymes catalase and superoxide dismutase that breakdown these molecules.

4- Micro-aerophilic bacteria: These require for growth a low oxygen tension; lower than that present in the atmosphere: e.g. *Campylobacter jejuni*.

B- Carbon dioxide (CO₂) in minute quantities as that present in air is required by most bacteria. Certain species of bacteria, e.g. *Brucella abortus* and the pathogenic neisseria require higher concentrations (5-20%), which should be included in the atmosphere surrounding the culture.

6- Light:

Most bacteria are injured by diffuse day light and are killed within few hours by direct sunlight. Bright day light has a similar effect but milder. The action of day light and sunlight is due to the presence of ultraviolet rays which has a bactericidal action.

On the other hand, some bacteria must be exposed to a source of light since it is their source of energy. Those types of bacteria are saprophytic and contain chlorophylls. They are known as photosynthetic autotrophic organisms.

7- Bacterial Bi-products:

Bacterial reproduction is so rapid that the bacteria soon exhaust their food supply and excrete or release products that inhibit further bacterial growth. Acids are the most important bi-products according to their action on changing the PH of media.

In addition to the acids which affect the bacterial growth there are other bi-products of bacteria which may be used for identification of bacteria as pigments or products which can be used for production of vaccines against the dangerous effect of these microorganisms (toxoid).

Many bacteria are able to decompose fats which lead to the hydrolysis of fat into glycerol and fatty acids and then to simpler material.

The degree by which organism can act on proteins, fats and ferment carbohydrates can be used as a basis for identification and typing of different microorganism. The reaction is known as the biochemical reactions (Biotyping).

Viruses: They are very small particles, ultramicroscopic, so they are not seen by ordinary microscope. Measured by nanometer (nm)

Note: $1 \text{ nm} = 10^{-9} \text{ m}$

All viruses show an extreme degree of parasitism. They are dependent for their life and reproduction on living cells which they invade. They can infect animal cells, plant cells, even bacterial cells.

The main difference between bacteria and viruses are:

- 1- They are obligate intracellular parasites. They can be cultivated only on living tissue culture cells.
- 2- They are very minute; that they could only be seen by the electron microscope.
- 3- The viruses possess only one type of nucleic acid; either DNA or RNA.

Collection of specimens for bacteriological investigations

Some important precautions should be taken in collecting clinical specimens for bacteriological study:

- 1) The specimen should be collected from the actual diseased tissue.
- 2) The more material send for investigation, the greater the chance of isolating the causative organism.
- 3) The Specimens must be collected before the initiation of antibiotic treatment.
- 4) The samples should be put in a sterile container, securely closed, adequately labeled and dated.
- 5) Handle, store and ship specimens appropriately.
- 6) The accompanying sheet should include the patient's name, age, sex, location, diagnosis, relevant data (e.g. if there is any treatment with antibiotics or any chemotherapeutic agent).
- 7) All specimens for culture are dangerous to the nurses and laboratory staff who handle them so Personal Protective Equipment (PPE) as Gloves, Mask, Protective clothing should be used.

Nasal swab

A dry swab is inserted 2 cm into the nostril and left in place for a few seconds to absorb sample material. It is then slowly withdrawn with a rotating motion. Specimens from both nostrils are obtained with the same swab. The tip of the swab is put into a sterile container.

Nasopharyngeal swab

A flexible, fine-shafted swab is inserted into the nostril and back to the nasopharynx and left in place for a few seconds. It is then slowly withdrawn with a rotating motion. A second swab should be used for the second nostril. The tip of the swab is put into a sterile container.

Throat swab:

Use tongue depressor to hold the tongue down, rub the swab over the posterior pharynx and tonsillar area then withdraw the swab being careful not to touch cheeks, teeth or gums.

Ear swab:

Not use antibiotic at least 3 hours before ear swab, Place a sterile swab into the outer ear and gently rotate to collect the secretions, for deeper ear swabbing a speculum may be used but only experienced medical staff should undertake this procedure as damage to the eardrum may occur.

Eye swab:

Ask the patient to look upwards and gently pull the lower lid down. Use a sterile swab and gently roll the swab over the conjunctival sac inside the lower lid.

Sputum sample:

It's best to collect sputum specimens first thing in the morning. They should be instructed to sit up and take several deep breaths to loosen secretions before giving a deep cough to release the sputum. Check that the sample contains sputum, as samples contaminated with saliva are difficult to interpret and can be misleading

It is important to note that droplets and aerosols may be generated when collecting sputum specimens, so health professionals should use PPE.

In patients suspected to have Tuberculosis take 3 specimens on 3 consecutive days as the tubercle bacilli may be present in few numbers in the sputum.

Urine sample:

Early morning specimen of urine is recommended (accumulated in the bladder overnight). After Cleansing of the skin surrounding the urethral meatus to avoid contamination of the specimen with organisms normally present on the skin. Specimens of urine should be (mid stream specimens) collected into a wide sterile specimen container.

Stool sample:

Ask the patient to urinate first to prevent contamination of the stool with urine.

Place something in the toilet to catch the stool, such as a potty, empty plastic container or plastic wrap over the rim of the toilet.

Make sure the stool doesn't touch the inside of the toilet.

Use the spoon that comes with the container to collect the stool, and then screw the container.

Put anything you used to collect the stool in a plastic bag, tie it up and put it in the bin.

Procedure for taking blood cultures:

Prior to venipuncture the skin must then be decontaminated with a 70% alcohol and allowed to dry. Do not re-palpate vein (even with a gloved hand) after decontamination.

Use both blood culture bottles (aerobic and anaerobic), remove the plastic cap and scrub the bung with 70% alcohol wipe for 15 seconds and allow to dry prior to inoculation.

After withdrawing the blood, and insert the blood into the culture bottles remove and safely discard the needle in safety box.

When inoculating the blood culture bottles, the anaerobic culture bottle should be inoculated first and then the aerobic culture

bottle, so that oxygen trapped in the syringe will not be transferred to the anaerobic bottle.

The specimen should preferably be taken during fever episodes as more bacteria may be present at that time.

Wound swabs

Obtain the specimen prior to any dressing or cleaning procedure of the wound. This will maximise the material obtained and prevent killing of the organism by the use of antiseptics. .

Use a sterile swab and gently rotate on the area to collect exudate from the wound and place into transport medium.

Where there is pus collect as much as possible in a sterile syringe and put in a plain sterile container (do not use a swab) and send to the laboratory.

Serum sample:

Serum is the liquid fraction of whole blood that is collected after the blood is allowed to clot.

It is the usual procedure to take blood at the beginning of a disease and then again after 10 because an increase in the amount of antibodies in the second specimen will be an indication of continued stimulation by the organism which must therefore be present in the patient.